



Document Management System



Sultanate of Oman

Ministry of Health

The Royal Hospital

Department of Child Health

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Title: Severe / Life Threatening Asthma

Definition:

- Asthma is life threatening when associated withProgressive respiratory failure (cyanosis, exhaustion and confusion/drowsiness)and/or failure to respond to maximal bronchodilators.
- The need for ICU admission should be decided onclinical state and response to treatment and not by blood gases.
- Admission fromHigh dependency to PICU is appropriate if there is life threatening asthma orthere is a failure to respond to maximal therapy by 6 hours. The PICU registrarshould be called to see all children prior to starting an IV salbutamolinfusion in HD.
- No infant (<1 year) should be started on intravenous bronchodilators without discussionwith a consultant.

Arterial blood gas:

- Early phase: hypoxemia, hypocarbia
- Impending respiratory failure: hypercabia

Management

1. 100%Oxygen. High flow (>10 l/m) vianon-rebreathing mask.
2. Continuousnebulized Salbutamol. 5mg/dose for all ages.
3. Ipratropiumbromide250mcg/ml. Add 1ml to the salbutamol, repeat every 20min for 3doses,then every 4 hours.
4. Methylprednisolone1mg/kg 6hrly IV for 24 hours, then 12hrly for 24 hours then daily.
5. IV magnesiumsulphate (50%). Give 0.1ml/kg (approx. 50mg/kg) in 20mls 5% dextrose over 20m,maximum dose 5ml. This may be repeated 1-2 times every 6 hours.
6. RegularMagnesium Sulphate; 50mg/kg 6 hourly or Magnesium infusion 0.06ml/kg/hr. keepserum Mg level 1.5-2.5mmol/L.

Continuous Aminophylline infusion according to child age:

- **Aged 1-9 years:** 1.1mg/kg/hr. (55mg/kg Aminophylline in 50ml 5% dextrose at 1ml/hr)
- **Aged ≥ 10 years, <35 kg:** 0.7mg/kg/hr.(35mg/kg Aminophylline in 50ml 5% dextrose at 1 ml/hr)
- **Aged ≥ 10 years, >35 kg:** 0.7mg/kg/hr.(neat Aminophylline (25mg/ml) at 0.028ml/kg/hr)

8. IV Salbutamol; 5mcg/kg/min over 1 hour and then 1mcg/kg/min. Check for hypokalaemia and lactic acidosis 6 hourly. Give in addition to nebulized Salbutamol.

9. **If continued deterioration despite above measures:**

· Non Invasive ventilation: Consider NIV early on. NIV can be tried if child failed to respond to Magnesium sulfate.

· Intubation and ventilation. Always ensure PICU consultant has been called.

· Basic principles:

- Use cuffed tube

- Use pressure control ventilation

- IT: 0.8sec

- PEEP <5 or $<$ auto PEEP

- Ventilate with low rate, so that expiration completed before next breath starts if possible

- Acceptable breath Rates: 1-9 yrs. 15-20/min. >10 yrs. 6-15/min

- Accept PaCO_2 60-80mmHg, pH 7.20-7.25, provided hemo-dynamically stable

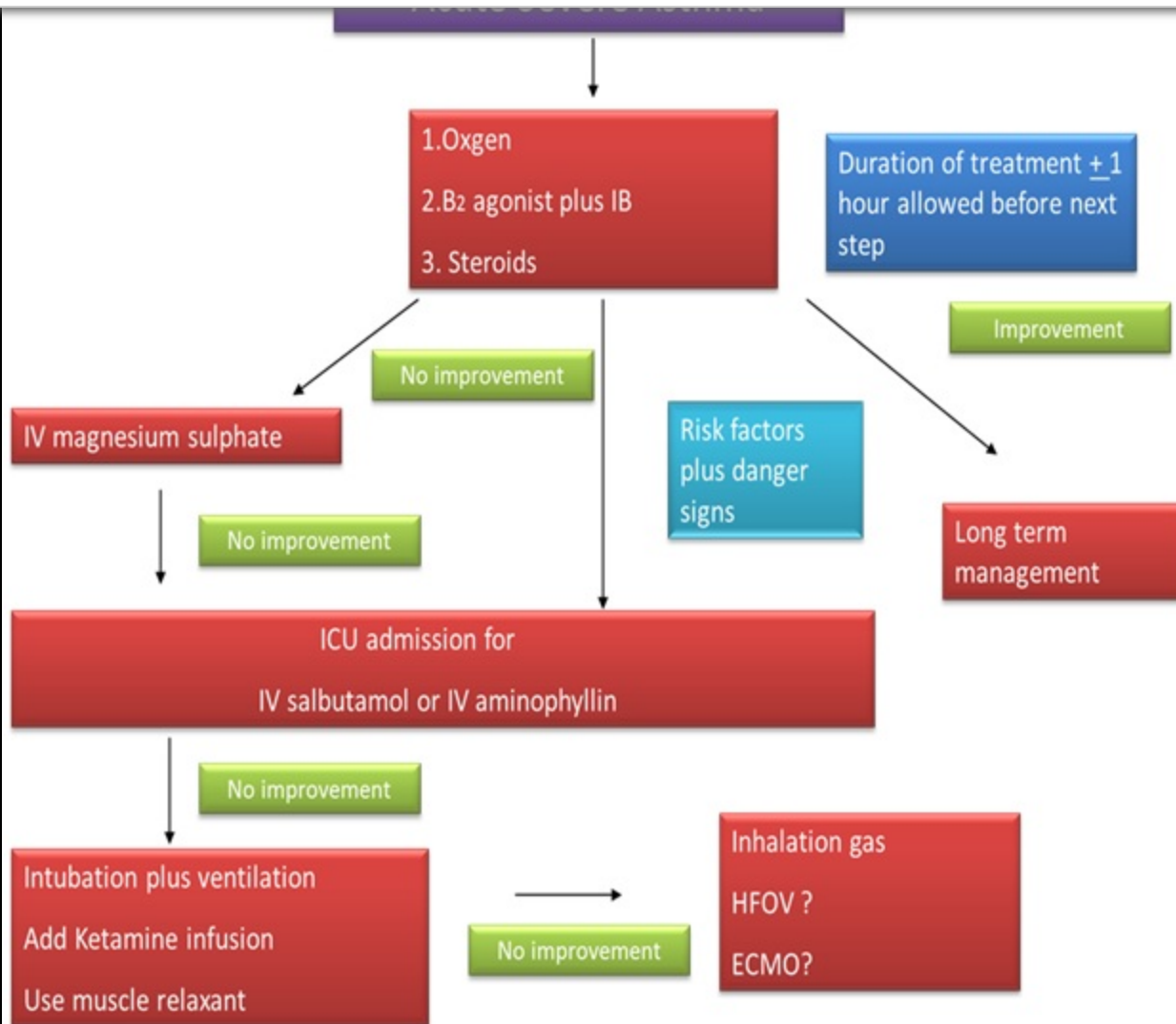
- Use ketamine for sedation 10- 20mcg/kg/min.

- Use muscle relaxant if required

o **If the child hypotensive or has low central venous pressure (CVP) give 10 ml/kg of 0.9% saline as bolus but avoid vigorous hydration**

Once improving

1. Wean off Salbutamol infusion.
2. Wean off aminophylline infusion.
3. Wean off Magnesium infusion
4. Reduce Salbutamol nebulizer from continuous to intermittent with progressive reduction in frequency
5. Change steroids to oral



References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
British guideline on the management of asthma	British thoracic society	Oct 2014	92-100

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